Q1 **Palindrome number/string** 10 marks

Palindrome number/string

Your Code

```
import java.util.*;
import java.lang.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int temp=n;
        int dig,rev=0,i;
        while(n!=0){
            dig=n%10;
            rev=rev*10+dig;
            n/=10;
        }
        if(temp==rev){
            System.out.println("palindrome num");
        }
        else{
            System.out.println("not a palindrome num");
        }
        sc.nextLine();
        String s=sc.nextLine();
        String t=s;
```

Input

```
121
cac
```

Output

Q2 **First n perfect numbers** 10 marks

First n perfect numbers

Your Code

```
import java.util.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int i,j;
        for(i=1;i<n;i++){
            int sum=0;
            for(j=1;j<i;j++){
                if(i%j==0){
                    sum+=j;
                }
            }
             if(sum==i){
                    System.out.println(j+" ");
                }
            }
        sc.close();
    }
}
```

Input

```
500
```

Output

```
6
28
496
```

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
import java.util.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int []a=new int[n];
        int i,j,temp;
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int nmax=sc.nextInt();
        int nmin=sc.nextInt();
        if(nmin<n || nmax<n){
            for(i=0;i<n;i++){
                for(j=i+1;j<n;j++){
                    if (a[i]>a[j]){
                        temp=a[i];
                        a[i]=a[j];
                        a[j]=temp;
                    }
                }
            }
        }
```

Input

```
5
2
2
```

Output

```
2 largest num: 4
2 smallest num: 2
```

Decimal number equivalent to binary number and octal number

Your Code

```
import java.util.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int d=sc.nextInt();
        int de=d;
        int bin=0,base=1,rem;
        int oc=0,b1=1,r1;
        while(d!=0){
            rem=d%2;
            bin=bin+(rem*base);
            base*=10;
            d/=2;
        }
        while(de!=0){
            r1=de%8;
            oc=oc+(r1*b1);
            b1*=10;
            de/=8;
        }
        System.out.println(bin);
        System.out.print(oc);
        sc.close();
```

Input

```
10
```

Output

```
1010
12
```

Find the average positive and negative numbers upto -1

Your Code

```
import java.util.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        double pavg=0,navg=0;
        int psum=0,nsum=0,c1=0,c2=0;
        int i;
        while(true){
            int n=sc.nextInt();
            if(n==-1){
                break;
            }
            else if(n>0){
                psum+=n;
                c1++;
            }
            else{
                nsum+=n;
                c2++;
            }
        }
            pavg=psum/c1;
            navg=nsum/c2;
```

Input

```
1
2
3
-5
```

Output

```
positive nos avg: 4.0
negative nos avg: -7.0
```

Q6 Removing duplicates in array 10 marks

Removing duplicates in array

Your Code

```
import java.util.*;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        Integer a[]=new Integer[n];
        int i;
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        HashSet<Integer> set=new HashSet<>(Arrays.asList(a));
        System.out.print(set);
        sc.close();
    }
}
```

Input

```
7
1
2
3
```

Output

```
[1, 2, 3, 4]
```

Run Save

Matrix multiplication

Your Code

```
import java.util.Scanner;
public class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int r1=sc.nextInt();
        int c1=sc.nextInt();
        int r2=sc.nextInt();
        int c2=sc.nextInt();
        int [][]a=new int[r1][c1];
        int [][]b=new int[r2][c2];
        int [][]c=new int[r1][c2];
        int i,j,k;
        for(i=0;i<r1;i++){
            for(j=0;j<c1;j++){
                a[i][j]=sc.nextInt();
            }
        }
        for(i=0;i<r2;i++){
            for(j=0;j<c2;j++){
                 b[i][j]=sc.nextInt();
            }
        }
        if(c1!=r2){
            System.out.println("matrix multiplication not possible");
```

Input

```
2 2
2 2
1 2
3 4
```

Output

```
13 12
31 30
```